Examiner only

Option A – **Alternating Currents**

6. *(a)* The sinusoidal emf produced by a rotating coil in a magnetic field has a **peak** emf of 18.0 V and a frequency of 70.0 Hz. The coil is connected to the y-input of an oscilloscope.

(i) The output of the rotating coil is used to power a small lamp of known resistance. Explain how you would calculate the rms input power to the lamp from the known resistance and **peak** emf. [2]

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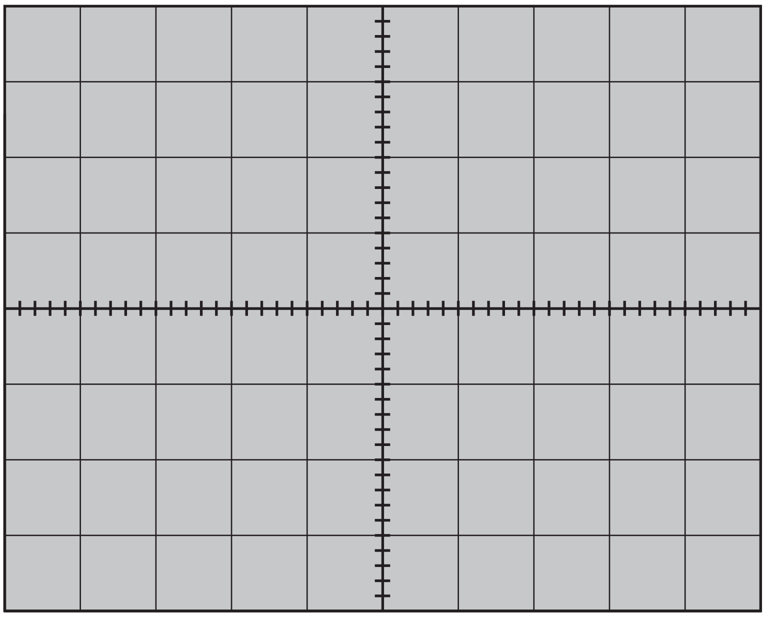
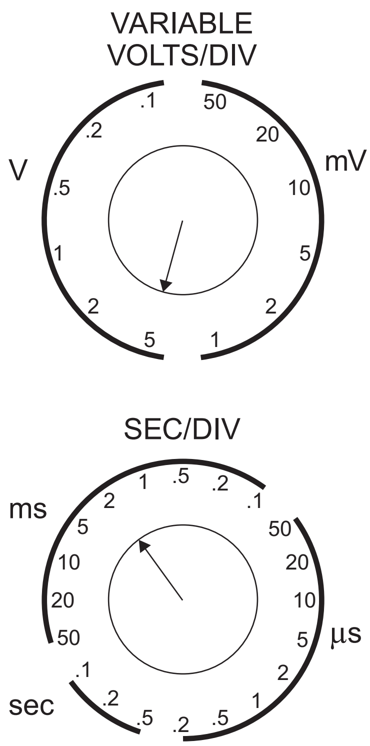
(ii) Draw a typical trace that might be seen on the oscilloscope screen with the oscilloscope settings shown. [5]

Space for calculations.





 (1420U40-1) **Turn over.**

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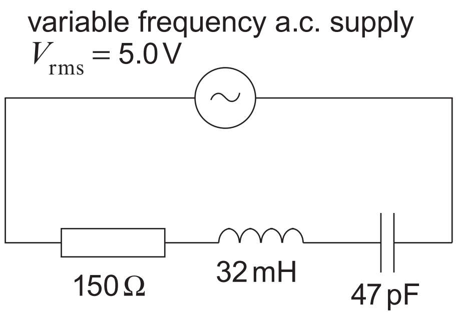
(b) An LCR circuit is shown below.



(i) Explain why the resonance frequency of the circuit occurs when: [3]

$$\omega L = \frac{1}{\omega C}$$

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(ii) Calculate the resonance frequency of the circuit. [2]

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(iii) Calculate the rms current when the frequency of the supply is 324 kHz. [3]

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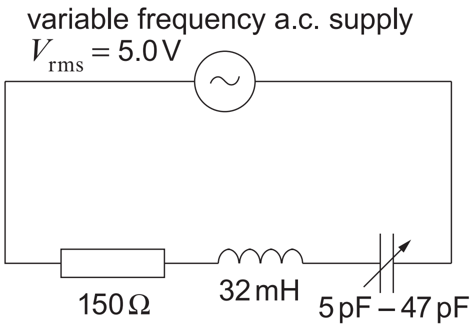
(iv) A student claims that the following circuit cannot have a peak pd above 1.5 kV across the capacitor. Investigate whether or not he is correct. [5]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$.)



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